SQL, a query language for relational databases

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Computational Management of Data – Part II (A.Y. 2025/2026)
Second Cycle Degree in Digital Humanities and Digital Knowledge
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Summary of the previous lectures (1/3)

A datum is a declarative statement subject-predicate-object that, through the predicate, either **attributes** a **literal** (i.e. a value such as a string, a number, etc.) to a subject **entity** *or* it **relates** such a subject **entity** with another **entity**

Each entity, being used either as subject or object of a statement, is characterised by **a unique identifier**

The **same entity** can be used as subject or object in one or more data, while a literal **cannot be used** as subject in any datum

An attribute is intrinsically **part of** the entity to which it is associated – modifying the value of an attribute affect **only** the entity to which it refers to

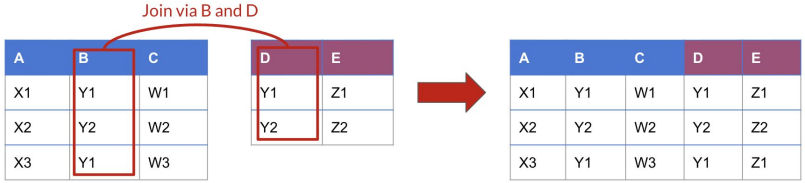
A **data model** is an abstract, simplified and formal representation of some data related to a system or a real domain, and enables us to describe what a data collection is about and to check data correctness

A **data model** permit one to specify <u>classes of entities, their attributes and relations</u>

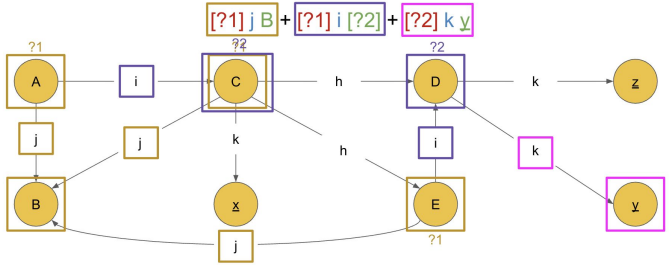
Summary of the previous lectures (2/3)

Depending on the structure in which data are stored , queries to datasets are approached differently:

- With **tabular data**, often you have to combine tables between them to obtain bigger tables which contain the query requirements and the related answer



- With **graph data**, you explore the graph starting from fixed points (i.e. known entities, values, predicates) to find a pattern that is compliant with the query



We use **descriptive statistics** as methods to summarize and describe data in a quantitative way – (measures of central tendency) Mean; Median; Mode, (measures of variability) minimum, maximum, and standard deviation

Summary of the previous lectures (3/3)

The creation of the database structures, the mechanism to upload new data into the database, and all the facilities for querying database data, are enabled by the **database management system (DBMS) –** available as **standalone** applications or Embedded DBMS

A **transaction** is a unit of work performed within a DBMS against a database – it represents any change in a database – are one of the main components that enables the integrity of data, being them executed fully or causing a rollback of the database

SQL

The Structured Query Language (SQL) is a query language used and designed for managing data in a relational database management system

It is a standard that is implemented in all relational database management systems, and allows one to create tables, to populate them, and finally to query them using a particular syntax

Be aware: even if it is a standard, porting SQL code from a database management system to another may require some changes

*A good tutorial is available at: https://www.sqlitetutorial.net/

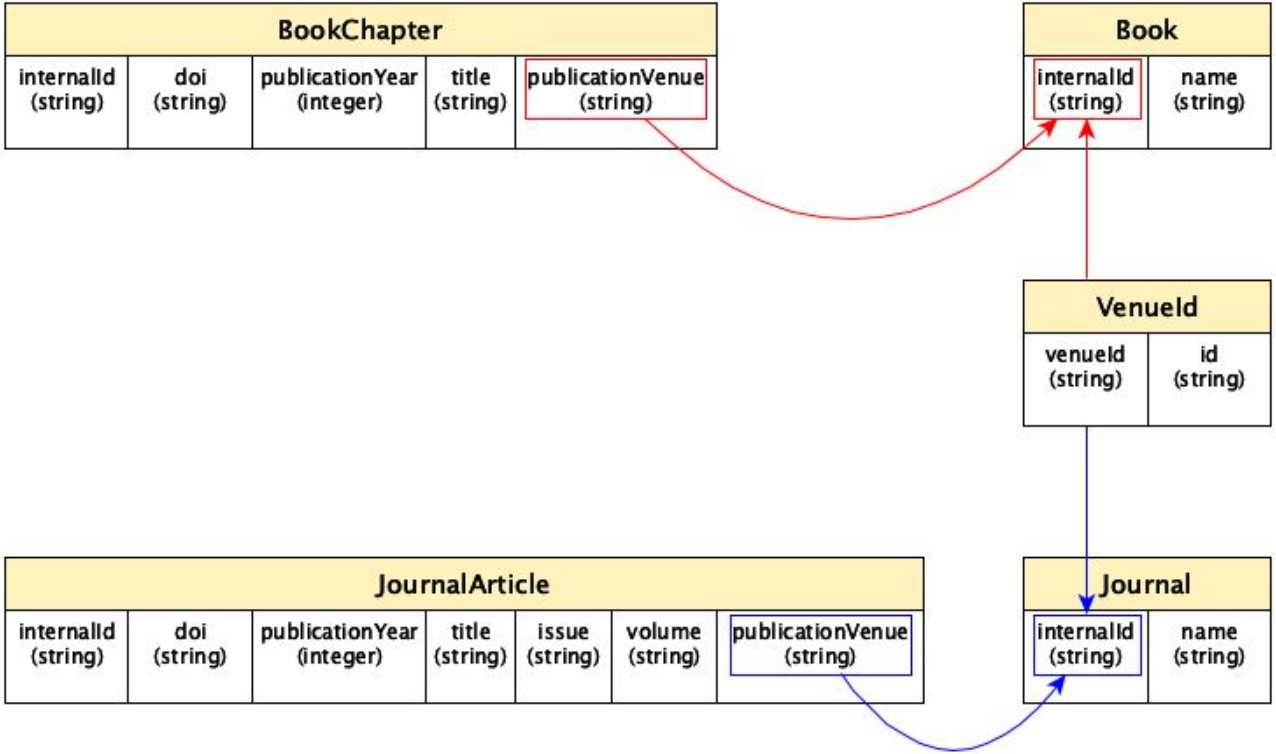
Basic SQL syntax for queries

`SELECT <columns>`	A comma-separated list of columns to have as result of the query ("*" means *all the columns*)
`FROM <tables>`	The table(s) from which to retrieve the data
`WHERE <conditions>`	*[OPTIONAL]* A condition that must hold to include the row of a table in the result of the query
`;`	

Questions to remind for designing a query:

- Which tables do you need?
- Which columns do you select?

Exemplar tables for queries



Data in the tables

VenueId

venueId	id
venue-0	1531-6912
venue-1	1367-5931
venue-2	9780470291092
venue-3	1027-3662

Book

internalId	name
venue-2	Proceedings of the 5th Annual Conference on Composites and Advanced Ceramic Materials: Ceramic Engineering and Science Proceedings

Journal

internalId	name
venue-0	Comparative and Functional Genomics
venue-1	Current Opinion in Chemical Biology
venue-3	Journal of Theoretical Medicine

BookChapter

internalId	doi	publicationYear	title	publicationVenue
publication-2	10.1002/9780470291092.ch20	1981	Mechanisms of Toughening in Ceramic Matrix Composites	venue-2

JournalArticle

internalId	doi	publicationYear	title	issue	volume	publicationVenue
publication-0	10.1002/cfg.304	2003	Development of Computational Tools for the Inference of Protein Interaction Specificity Rules and Functional Annotation Using Structural Information	4	4	venue-0
publication-1	10.1016/s1367-5931(02)00332-0	2002	In vitro selection as a powerful tool for the applied evolution of proteins and peptides	3	6	venue-1

Queries

(Q1) Retrieve complete information about all journal articles

(Q2) Retrieve the titles of all journal articles

(Q3) Retrieve the title of the journal article with DOI “10.1016/s1367-5931(02)00332-0”

(Q4) Retrieve the title of the publication with DOI “10.1016/s1367-5931(02)00332-0”

(Q5) Return the name of the journal of the article with DOI “10.1016/s1367-5931(02)00332-0”

(Q6) Return the id and name of the journal of the article with DOI “10.1016/s1367-5931(02)00332-0”

Q1

Retrieve complete information about all journal articles

- Which tables do you need? **JournalArticle**
- Which columns do you select? ***(all columns)**

```
SELECT *
FROM JournalArticle;
```

JournalArticle

internalId	doi	publicationYear	title	issue	volume	publicationVenue
publication-0	10.1002/cfg.304	2003	Development of Computational Tools for the Inference of Protein Interaction Specificity Rules and Functional Annotation Using Structural Information	4	4	venue-0
publication-1	10.1016/s1367-5931(02)00332-0	2002	In vitro selection as a powerful tool for the applied evolution of proteins and peptides	3	6	venue-1

Q2

Retrieve the titles of all journal articles

- Which tables do you need? **JournalArticle**
- Which columns do you select? **title**

```
SELECT title
FROM JournalArticle;
```

JournalArticle

internalId	doi	publicationYear	title	issue	volume	publicationVenue
publication-0	10.1002/cfg.304	2003	Development of Computational Tools for the Inference of Protein Interaction Specificity Rules and Functional Annotation Using Structural Information	4	4	venue-0
publication-1	10.1016/s1367-5931(02)00332-0	2002	In vitro selection as a powerful tool for the applied evolution of proteins and peptides	3	6	venue-1

Q3

Retrieve the title of the journal article with DOI “10.1016/s1367-5931(02)00332-0”

- Which tables do you need? **JournalArticle**
- Which columns do you select? **title**

```
SELECT title
FROM JournalArticle
WHERE
doi='10.1016/s1367-5931(02)00332-0';
```

JournalArticle

internalId	doi	publicationYear	title	issue	volume	publicationVenue
publication-0	10.1002/cfg.304	2003	Development of Computational Tools for the Inference of Protein Interaction Specificity Rules and Functional Annotation Using Structural Information	4	4	venue-0
publication-1	10.1016/s1367-5931(02)00332-0	2002	In vitro selection as a powerful tool for the applied evolution of proteins and peptides	3	6	venue-1

Q4

Retrieve the title of the publication with DOI “10.1016/s1367-5931(02)00332-0”

- Which tables do you need? **BookChapter** **JournalArticle**
- Which columns do you select? **title**

```
SELECT title
FROM BookChapter
WHERE
doi='10.1016/s1367-5931(02)00332-0'

UNION

SELECT title
FROM JournalArticle
WHERE
doi='10.1016/s1367-5931(02)00332-0';
```

BookChapter

internalId	doi	publicationYear	title	publicationVenue
publication-2	10.1002/9780470291092.ch20	1981	Mechanisms of Toughening in Ceramic Matrix Composites	venue-2

JournalArticle

internalId	doi	publicationYear	title	issue	volume	publicationVenue
publication-0	10.1002/cfg.304	2003	Development of Computational Tools for the Inference of …	4	4	venue-0
publication-1	10.1016/s1367-5931(02)00332-0	2002	In vitro selection as a powerful tool for the applied evolution of …	3	6	venue-1

Q5

Return the name of the journal of the article with DOI "10.1016/s1367-5931(02)00332-0"

- Which tables do you need? **JournalArticle** **Journal**
- Which columns do you select? **name**

```
SELECT name
FROM JournalArticle LEFT JOIN Journal ON
     JournalArticle.publicationVenue
     ==
     Journal.internalId
WHERE doi='10.1016/s1367-5931(02)00332-0';
```

Journal

internalId	name
venue-0	Comparative and Functional Genomics
venue-1	Current Opinion in Chemical Biology
venue-3	Journal of Theoretical Medicine

JournalArticle

internalId	doi	publicationYear	title	issue	volume	publicationVenue
publication-0	10.1002/cfg.304	2003	Development of Computational …	4	4	venue-0
publication-1	10.1016/s1367-5931(02)00332-0	2002	In vitro selection as a powerful tool for the applied evolution of …	3	6	venue-1

JournalArticle ⋈ Journal

internalId	doi	publication Year	title	issue	volume	publication Venue	internalId	name
publication-0	10.1002/cfg.304	2003	Dev…	4	4	venue-0	venue-0	Comparative and …
publication-1	10.1016/s1367-5931(02)00332-0	2002	In…	3	6	venue-1	venue-1	Current Opinion in Chemical Biology

Q6

```
SELECT id , name
FROM JournalArticle LEFT JOIN Journal ON
     JournalArticle.publicationVenue == Journal.internalId
     LEFT JOIN VenueId ON Journal.internalId == VenueId.venueId
WHERE doi='10.1016/s1367-5931(02)00332-0';
```

Return the id and name of the journal of the article with DOI "10.1016/s1367-5931(02)00332-0"

- Which tables do you need? **JournalArticle** **Journal** **VenueId**
- Which columns do you select? **id** **name**

Journal

internalId	name
venue-0	Comparative and Functional Genomics
venue-1	Current Opinion in Chemical Biology
venue-3	Journal of Theoretical Medicine

JournalArticle

internalId	doi	publicationYear	title	issue	volume	publicationVenue
publication-0	10.1002/cfg.304	2003	Dev...	4	4	venue-0
publication-1	10.1016/s1367-5931(02)00332-0	2002	In...	3	6	venue-1

VenueId

venueId	id
venue-0	1531-6912
venue-1	1367-5931
venue-2	9780470291092
venue-3	1027-3662

JournalArticle ⋈ Journal ⋈ VenueId

internalId	doi	pub…	title	issue	volume	pub…	internalId	name	venueId	id
publication-0	10.1002/cfg.304	2003	De…	4	4	venue-0	venue-0	Comp…	venue-0	1531-6912
publication-1	10.1016/s1367-5931(02)00332-0	2002	In…	3	6	venue-1	venue-1	Curr…	venue-1	1367-5931

END

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